SPARKLING REVOLUTION

A CHEAPER, GREENER ALTERNATIVE TO MINED DIAMONDS, THE LAB GROWN DIAMOND IS EXCITING DIAMANTAIRES ACROSS THE WORLD AND IN INDIA. GOVERNMENT PUSH AND THE ENTRY OF NEW PLAYERS OPEN UP A BRIGHT UNIVERSE OF PROFITABLE OPPORTUNITY

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Deep inside a two-storeyed building, overlooking a well-manicured lawn at a diamond processing unit in Surat in Gujarat, sits an air-conditioned laboratory where a curious process is at work. In a few dozen non-descript machines, each the size of a small refrigerator, being meticulously monitored digitally each second by an expert, is growing a fruit like no other: a diamond in the rough.

A little further, in a building just across the lawn, dozens of polishers seated in another air-conditioned room are focused on the stones in their hands, illuminated by powerful table lamps. Classic Kishore Kumar songs playing in the background, they deftly chip and shape the tiny stone into a 'princess-cut' diamond. There was a time when these precious gems would travel miles across continents—from the African mines to the diamond bourses in Europe—before landing in the hands of skilled cutters and polishers in India. Today, what you see 60-year-old Ramesh Sondakar handling, for instance, has just come from the opposite building.

Lest you think you are on the precincts of a fake diamond factory, think again. You are in the presence of what are rather unpretentiously called lab grown or synthetic or man-made diamonds.

WHAT IS A LAB DIAMOND?

An American diamond by another name, the cubic crystalline form of zirconium dioxide? Far from it. A diamond is essentially, as the US Federal Trade Commission defines it, "pure carbon crystallised in the isometric cubic system". A lab-grown diamond (LGD) mimics the natural process of diamond formation but in a controlled laboratory environment. It has the same chemical, physical and optical properties as its naturally occurring counterpart. Put it alongside a mined diamond, and it is impossible to tell the difference, something that Sondakar, with his years of experience, admits he cannot do either, even under the most powerful lens. It is only a certification by the Gemological Institute of America (GIA) or the International Gemological Institute (IGI) that can tell you if you are buying a mined diamond or a synthetic one. The latter is assessed on the same '4Cs'—colour, cut, clarity and carat weight—that are used to evaluate naturally occurring diamonds. Available